

Project Proposal: Disaster Response Tweet Classification

Course: CSE 4122 (Natural Language Processing Laboratory)

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1. Project Overview

During natural disasters (e.g., earthquakes, floods, hurricanes), social media platforms like Twitter produce critical real-time information. However, emergency responders face severe information overload mixed with noise. This project aims to build an end-to-end multiclass NLP classification pipeline to categorize disaster-related tweets into actionable humanitarian classes, enabling rapid situational awareness and efficient resource dispatch.

2. Dataset & Classes

The project utilizes the HumAID dataset, comprising over 77,000 human-annotated disaster-related tweets across 11 target classes.

#	Class Name	Example Tweet
1	Caution and Advice	<i>"Tsunami warning active. Move to high ground immediately."</i>
2	Displaced People and Evacuations	<i>"Mandatory evacuation order for Zone A; shelter at civic center."</i>
3	Infrastructure and Utility Damage	<i>"Highway bridge collapsed; major power outage across downtown."</i>
4	Injured or Dead People	<i>"Officials confirm 12 dead and over 50 injured in the blast."</i>
5	Missing or Found People	<i>"Looking for John Doe, missing near Sector 4 flood area."</i>
6	Requests or Urgent Needs	<i>"Trapped on roof, need drinkable water and medical supplies asap."</i>
7	Rescue, Volunteering, or Donation	<i>"Red Cross deployed rescue boats to evacuate stranded families."</i>
8	Sympathy and Support	<i>"Praying for everyone affected by the earthquake. Stay safe!"</i>
9	Other Relevant Information	<i>"Meteorology center says storm will make landfall in 2 hours."</i>
10	Not Humanitarian	<i>"Watching storm coverage on TV while having a hot coffee."</i>
11	Don't Know / Can't Judge	<i>"Did anyone just hear a loud rumble outside?"</i>

3. Proposed Methodology

- **Preprocessing:** Lowercasing, URL/mention stripping, hashtag segmentation, and tokenization.
- **Model Development:** Evaluating baseline ML models (TF-IDF with Logistic Regression, SVM) alongside fine-tuned pretrained Transformers (BERT, RoBERTa) using weighted Cross-Entropy loss.
- **Evaluation Metrics:** Macro/Micro F1-Score, Class-wise Precision, Recall, and Confusion Matrix analysis.